Do Greater Sanctions Deter Youth Crime? Evidence from a Regression Discontinuity Design

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Abstract

We exploit the discontinuous jump in criminal sanctions at the age of majority in conjunction with administrative data from California to generate regression discontinuity estimates of the deterrent effect. Estimates show that the greater severity imposed upon adolescents at age 18 deters violent crime by 10-12%. Results are robust to multiple techniques and specifications. Using these results, we estimate an elasticity of crime with respect to sanction intensity that ranges from -0.145 to -0.174. We extend our results to demographic sub-populations and find female offenders, as well as white and Asian offenders, are relatively more responsive to sanctions.

Keywords: Juvenile crime, deterrence, regression discontinuity.

JEL Classification: I38, J13, K42

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1 Introduction

Crime remains a tremendously costly problem across human societies. As recently as 2012, crime victims in the US suffered injury and loss of life totaling over $750 billion. Meanwhile, crime control expenditures exceeded $646 billion. Hence, policy decisions that impact crime control and mitigation are high stakes matters with the potential for enormous societal gains and losses. Moreover, there is considerable debate in US policy circles. Some are strongly in favor of reduced sentencing, while others are not. For compelling reasons criminal deterrence is one particular component of crime control that is of tremendous interest. Put simply, crimes that are prevented through deterrence prevent both the creation of victims, and the high costs associated with punishing and rehabilitating offenders.

Economic theory has long held that deterrence is a very real phenomenon in that rational agents reliably change their behavior in the face of altered incentive structures. The canonical model of Becker (1968) posits that agents choose an optimal level of crime by weighing the expected benefits of crime against the expected costs where expected costs are a function of the probability of apprehension and the punishments that follow apprehension. Many policies that intend to reduce crime take the form of increased sanctions that induce rational agents to reduce criminal activity by virtue of a deterrent effect. The premise of a deterrent effect as a theoretical matter is widely accepted and non-controversial.

Despite the enormous desire to deter crime, empirical evidence for deterrence stemming from criminal sanction policies has been rather mixed; well designed and implemented studies have reached different conclusions. For example, Hjalmarsson (2009), Iyengar (2008), Loeffler and Chalfin (2017) and Lee and McCrary (2017) find that an increase in sanction intensity does not result in meaningful deterrence, while Levitt (1998), Helland and Tabarrok (2007), Drago, Galbiati, and Vertova (2009) and Hansen (2015) find evidence of more substantial deterrence effects. As such, there has not been a convergence to a widely-accepted consensus regarding the existence or magnitude of deterrence effects stemming from contemporary sanction policies.

In the spirit of Lee and McCrary (2017) and Loeffler and Chalfin (2017), we consider the discontinuous and quasi-random change in exposure to penalties that occurs at the age

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of majority, but do so in the state of California. Prior to the age of majority, offenders face penalties within the purview of juvenile justice systems. Following the age of majority, offenders are thrust into adult correctional systems which are characterized by harsher punishments. Conventional theory unambiguously points to an immediate decline in criminal acts as a result of deterrence. To test this assertion we use high frequency administrative data that spans the universe of both adult and juvenile arrests, as well as the precise age of all offenders at the time of arrest. To exploit the discontinuous jump in penalties and uncover the deterrence effect in isolation we utilize a regression discontinuity (RD) design using both global parametric and local nonparametric approaches where the running variable is the offender's age to the day. Note that at an age of 17 years 364 days offenders face juvenile penalties, while a day later at the age of 18, they face adult sanctions. Our RD results suggest that the greater punitiveness of the adult corrections system does indeed have substantial deterrence effects. We find that adolescents are highly responsive in reducing violent criminal acts, but find weaker evidence that they respond by altering property crimes. On average we find that violent crime drops by about 10-12% and property crimes drop by up to 8%. Our findings are highly significant at both statistical and practical levels. Moreover, our results are robust to a variety of changes in specification, estimation technique and inclusion of controls. We consider multiple crime categories and consider a multitude of different specifications using both the global parametric approach to RD and the local nonparametric approach.

Our results contribute to the literature in at least three ways. First, we find broad support for a theory of rational offenders that reduce criminality in the face of increased criminal sanctions. Secondly, we offer strong evidence that deterrence effects exist specifically for adolescents, and that youth are responsive to the threat of greater sanctions. Finally, we find considerable evidence for heterogenous deterrence effects that vary by race and gender.

We argue that our estimates of the treatment effect from a regression discontinuity are attractive for a variety of reasons. Namely, the identifying assumptions are fairly weak. The only required assumptions are that criminal trends around the discontinuity are smooth in the absence of exposure to the treatment (a greater degree of punitiveness following the age of majority) and that youth cannot effectively manipulate or misrepresent their birth dates.

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However, RD estimates will not reveal the time-varying nature of offending around the age of majority. As such, we also employ a nonparametric event study framework that allows us to examine the time-varying pattern of effects. We find evidence that deterrence effects are persistent, and not merely transitional.

In order to interpret our main results as an elasticity of crime with respect to sanction intensity, we construct a measure of the difference in punitiveness between the juvenile justice system and the adult justice system. Doing so in conjunction with our estimates of the reduction in crime as a result of the greater severity of adult corrections yields an elasticity of roughly -0.145 to -0.174. These estimates place our results between other commonly cited measures of elasticity within the extant literature.

The rest of the paper is as follows. In Section 2 we provide the necessary background information, in Section 3 we introduce our data and methodology, Section 4 presents our main results, Section 5 discusses robustness and threats to identification, Section 6 considers extensions and presents additional results, and Section 7 concludes.

2 Background

While Bentham (1789) and Beccaria (1764) were perhaps the first to rigorously discuss deterrence, Becker (1968) developed the pioneering framework that describes crime as most economists now view the matter. Agents in the Becker model are rational and view criminal acts as potentially attractive gambles that are selected when the expected utility of crime exceeds the expected utility of abstaining from crime (usually viewed as the wage returns for legal work). The expected utility of a criminal act depends upon: (1) the utility from committing a crime and escaping apprehension, (2) the utility from committing a crime followed apprehension and punishment, and (3) the probability of apprehension. Deterrence occurs when rational offenders choose to commit fewer criminal acts as a result of changed incentives that can result from either increases in the likelihood of apprehension, or increases in the severity of punishment.

An extensive literature looks at the value of increasing the probability of apprehension as a means of deterrence. Influential papers are many, but include, for example, Levitt (1997), Di

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Tella and Schargrodsky (2004), and Draca, Machin, and Witt (2011).[1] This paper concerns itself with changes in the punishments or sanctions an offender faces conditional on being apprehended. From a policy standpoint, deterrence through changes in criminal sanctions may be of particular interest since policy makers have considerable latitude in determining the severity of sanctions and society's choice of sanction regime can be made both public, and fairly transparent. In contrast, the returns to abstention from crime and legal work, are presumably less under the purview of policy makers. Additionally, altering the probability of apprehension may be considerably more difficult than altering punishments, while also being less easily communicated to the public.

2.1 Juvenile versus Adult Corrections

Our estimates of deterrence effects rely on the fact that adolescents that are adjudicated in the juvenile justice system are exposed to substantially less punitive measures than nearly identical adolescents that are adjudicated in the adult justice system by virtue of having reached the age of majority at the time of their offense. The large differences between the two justice systems stem from meaningful differences in philosophies and practices. Philosophical differences primarily focus on juveniles' greater malleability and potential for correction. Minors are seen as not fully formed emotionally, mentally and physically, and therefore the juvenile justice system seeks to alter juvenile maturation in positive ways that will decrease the probability of recidivism. As such, juvenile justice systems tend to emphasize the correctional nature of the justice system and relatively de-emphasize the penal aspect of the justice system.

In practice these philosophical differences mean that juveniles are far more likely to receive probation, counseling, education and rehabilitation than adults. Additionally juveniles are far more likely to be released following arrest than adults. Furthermore, juveniles are less likely to be incarcerated; and when they are, the sentence tends to be shorter than adult sentences. Lee and McCrary (2017) document this and show that the expected sentence duration for an adult is greater than double the expected sentence length for juveniles.

[1] Another extensive literature considers the effect of changing returns to legal work on crime, including for example Ehrlich (1973), Grogger (1998), Lochner and Moretti (2004), and Machin and Meghir (2004).

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Finally, in a qualitative sense the juvenile incarceration environment tends to be less punitive. Juvenile detention facilities generally have proportionally greater staffing levels and greater time allocation devoted to education and rehabilitation than adult facilities. Hence, juvenile justice outcomes are substantially less punitive along numerous dimensions than adult justice outcomes. Additionally, within the context of California, Krisberg, Hartney, and Marchionna (2010) note that California's adult criminal justice system is indeed far more stringent than the juvenile justice system.

If potential offenders are aware of the fact that they face harsher punishments upon reaching the age of majority it is reasonable to conclude that this is very likely to induce behavioral changes. However, if it were the case that juveniles were unaware of changes in the punishments they may face, it would also be reasonable to conclude that juveniles may not alter their behavior. As shown in Lochner (2007) and Hjalmarsson (2009) evidence supports the assertion that adolescents are well aware of the change in the correctional system and sanctions at age 18. Interestingly, Glassner et al. (1983) conducts interviews with juveniles and young adults in New York that suggests a decline in criminal activity at the age of majority (16 at this time in New York) specifically because of well understood increases in punishments. To quote Glassner et al. (1983), and as it is reported in Levitt (1998), one individual reports upon reaching the age of majority: "When you are a boy, you can be put into a detention home. But you can go to jail now. Jail ain't no place to be." Another youth currently serving time in a juvenile detention center felt his current sentence was: "...easy. I'll just do my year and get out, it ain't nothing." Despite his unrepentant attitude he counseled his 16 year old friend to discontinue criminal acts: "Because he's 16 now; he'll go to jail."

3 Data and Methodology

3.1 Data

Our analysis relies most centrally on crime data from the state of California. Within California all law enforcement agencies are required to record all arrests and turn over that

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information to the California Department of Justice which compiles these administrative data into a census of all arrests in the state. This census of California arrests is known as the Monthly Arrest and Citation Register (MACR). In addition to recording all arrests for felonies and misdemeanors, the MACR includes important contextual information such as: the offender's date of birth, purported crime, offense level, race, gender, location of arrest, date of arrest, as well as other details.[2] The data is highly desirable for our purposes because among other things it spans all adults and all juveniles and it provides precise dates of birth and arrests. It also allows us to break down crimes by category, race, gender, and location. Our MACR data spans the period 1998-2006.

We organize the crime data in the following way. In each instance where an individual is arrested there is a corresponding observation in the data set. For each arrest we calculate the offender's exact age by combining the date of birth and the arrest date.[3] This allows us to construct daily aggregate arrest counts for all precise, daily ages.[4]

If there were differential record keeping for adults and minors, this would imply attenuation or downward bias in the estimate of any deterrence effect. This is because some juvenile offenses would be removed from the data and there would be an erroneous tendency to view crime as increasing at age 18 in spite of the presence of greater penalties. In the case of the MACR, any juvenile arrests are later hidden from courts following the individuals passage to the age of majority, but the juvenile record of an arrest remains in the MACR and is not removed. Hence, there is no differential record keeping between adults and juveniles in our study. Additionally, the 1994 Juvenile Justice Reform Act requires that records of all serious juvenile crimes be maintained and compiled as they have been in the MACR. Moreover, our central focus is on serious violent offenses that have not been purged from the MACR records.

In addition to measuring crime, we consider the observable characteristics of youth in their late teens to ensure that they display smooth trends at the age of majority for a variety of important life changes and events that may alter criminality. We use the American

[2]Racial determinations are made by the arresting law enforcement agencies.

[3]For example, if an individual is born on 01/01/1989 and is arrested on 12/01/2006, then the exact age at arrest is 17 years old and 335 days.

[4]For example, an aggregate crime count for 17 years and 47 days, and an aggregate crime count for 17 years 48 days and so on.

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Community Survey (ACS) and the Decennial Census to do so. The ACS is an annual survey conducted by the Census Bureau and provides a wealth of information at the individual level for both juveniles and adults across the entire country and California. In particular, it contains detailed information that allow us to evaluate potential threats to identification. We use the ACS and Decennial Census for all years in which data collection was conducted (2000-2006).

Finally, in order to frame our results as an elasticity we use adult incarceration data from the Census of State and Federal Adult Correctional Facilities, and juvenile incarceration data from the Census of Juveniles in Residential Placement. Both data sets are true censuses that account for all incarcerated youth and juveniles within California.

Table 1 presents summary statistics for adolescent and all criminal offenders. These descriptive statistics are presented for both violent and property crimes. Generally, violent offenders are more likely to be male and Hispanic, than female or other races within California. These trends are mirrored, to a slightly lesser degree, in the property crime categories. The bulk of both property and violent crimes occur in California's largest 18 counties. Additionally, consistent with prior research violent crime appears to peak later in the life cycle than do property crimes.

3.2 Empirical Strategy

We estimate the treatment effects at age 18 utilizing a regression discontinuity design. Specifically, we employ both a global parametric approach, as well as a local nonparametric approach in estimating deterrence effects. The global parametric approach is appealing in the sense that arguably, the data generation process of the conditional expectation function of crime over the life cycle can be closely approximated by a polynomial. If so, the estimates under the global parametric approach have higher power in fitting a smooth age-crime relationship as compared to alternative specifications. The regression equation under this approach takes the form:

$$Crime_i = \alpha + \beta\ \mathbf{1}\{a_i \geq 18\} + \sum_{k=1}^{K} \gamma_{0k}\ \mathbf{1}\{a_i < 18\}(a_i - 18)^k + \sum_{k=1}^{K} \gamma_{1k}\ \mathbf{1}\{a_i \geq 18\}(a_i - 18)^k + \epsilon_i \tag{1}$$

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Where $Crime_i$ is the frequency of crime for an individual at a certain age, a_i is the running variable age, with 18 being the age of majority. $\mathbf{1}\{a_i \geq 18\}$ and $\mathbf{1}\{a_i < 18\}$ are indicator functions that take value one when true. Parameter β measures the jump at the threshold — in this context, the deterrence effect at age 18. For this specification we focus on young adults between the ages of 16 to 19, inclusive.[5] This specification uses a Kth order polynomial with coefficients that differ on each side of the threshold. Based on recent work by Gelman and Imbens (2014), we estimate coefficients under $K = 1, 2, 3$. Reassuringly, our central results are highly consistent with respect to each other and display little variation under different choices of polynomial order.

We also perform the RD analysis utilizing local nonparametric techniques. Under the global parametric approach, estimators could potentially rely heavily on data far from the discontinuity (Gelman and Imbens (2014)). The local nonparametric approach is appealing in the sense that it overcomes this potential concern by assigning heavier weights to observations closer to the threshold. The standard triangular kernel is used for weight assignment due to optimality properties associated with boundary estimation problems as noted by Cheng (1997). This is summarized by the following set of equations:

$$\left(\hat{\beta}_1, \hat{\delta}_1\right) = \underset{\beta_1, \delta_1}{\arg\min} \sum_i \mathbf{1}\{a_i < 18\} K\left(\mathbf{1}\{|a_i - 18| \leq h\} \times \left(1 - \frac{|a_i - 18|}{h}\right)\right)\left(Crime_i - \beta_1 - \delta_1(a_i - 18)\right)^2 \tag{2}$$

$$\left(\hat{\beta}_2, \hat{\delta}_2\right) = \underset{\beta_2, \delta_2}{\arg\min} \sum_i \mathbf{1}\{a_i \geq 18\} K\left(\mathbf{1}\{|a_i - 18| \leq h\} \times \left(1 - \frac{|a_i - 18|}{h}\right)\right)\left(Crime_i - \beta_2 - \delta_2(a_i - 18)\right)^2 \tag{3}$$

Here $K(\cdot)$ is the triangular kernel function, and h is the bandwidth that vanishes to zero asymptotically. The treatment effect is characterized by the difference $(\beta_2 - \beta_1)$. For all RD analyses in this study we use the mean squared error minimizing bandwidth as put forth in Imbens and Kalyanaraman (2012). We also report the bias-corrected coefficients and standard errors to account for the fact that the Imbens-Kalyanaraman bandwidth is willing to accept to a certain extent the bias-variance tradeoff, and use robust standard errors to construct confidence intervals (as seen in Calonico, Cattaneo, and Titiunik (2014) and Calonico, Cattaneo, and Farrell (2017)).

[5] We have also utilized alternative age ranges. Our results are not sensitive to changes in the age range.

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3.3 Identifying Assumptions

In order for our methodology to reveal causal deterrence effects it must be the case that certain assumptions are satisfied and a variety of potential concerns are non-problematic. In this section we explain the identifying assumptions and consider their validity.

The primary identifying assumption is that in the absence of any changes induced by altered punishments, criminal behavior is fairly smooth and not discontinuous at age 18. Indeed, hypotheses about changes in criminal propensity are typically a function of gradual physical and psychological maturation that occurs in the late teens and early twenties. In addition, various life changing events such as high school graduation, moving out of the parents' home, beginning work or starting to live independently, may also have bearings on criminal tendencies. Thus, criminal behavior from ages 16-17, and 19-20 may be quite different. However, barring changes in sanctions, criminal tendencies evolve smoothly as maturation is a gradual process and major life events display smooth aggregate age effects as they occur at different points in time for different adolescents. Hence, use of annual crime data would likely be problematic in that criminality may not appear smooth across discrete ages.

We follow Grogger (1998), Levitt and Lochner (2001) and Lee and McCrary (2017) in solving this potential problem by using high frequency data. Our data reports arrests on a daily basis and allows for precise computation of the age of a criminal offender to uncover the discontinuous change in criminal offending at the age of majority. As such our smoothness assumption is unlikely to be violated. We show empirical evidence to further support the claim that the smoothness assumption is likely satisfied by using ACS and Decennial Census data to ensure that covariates are smooth around the discontinuity. This is discussed in greater detail in Section 5.

Another way the identifying assumption would be violated is if there exists systematic manipulation around the threshold that allows individuals to self select into, or out of, treatment. This is a non-problematic assumption in our study, given that criminals are unable to manipulate their date of birth. Note that the smoothness of the density of the RD running variable is frequently used as a test in a regression discontinuity designs, to ensure

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there is no manipulation around the threshold (McCrary (2008)). However, in this context, the sharp decline in density of criminal acts at the threshold is exactly the treatment effect of interest. As such, we note that we do not see any signs of misreporting dates of birth as there is no empirical evidence for a surge in criminal offenses immediately prior to the age of majority.[6]

Additionally, we establish the validity of our RD estimates by ruling out the possibility that any treatment effect is due to seasonality of births and criminal behavior. Suppose summer is when crime peaks, and people born in the fall are most inclined to engage in criminal activity due to lower educational attainment induced by mandatory kindergarten enrollment ages as noted by Angrist and Keueger (1991) and Cook and Kang (2016). Then these individuals could potentially engage in high levels of crime immediately prior to the age of majority, which for them would coincide with summer weather. Subsequently, criminal acts would then drop after the age of majority due to the shift to colder weather. Under these circumstances it would be possible to mistake seasonality effects for deterrence effects. If this concern were valid, then we would see discontinuous drops in crime every year immediately following passage to ages 16, 17, 18, 19 and 20. This is not the case in the data, rather there is *only* a discontinuous change at the 18th birthday.

4 Main Results

Table 2 presents the central results of our paper. Figure 1 shows an analogous visual representation of the discontinuity under the global parametric RD framework. In all specifications the treatment effect is the incidence of crime immediately at the threshold relative to crime below the threshold.

The dependent variable in all regressions is the count of arrests occurring a given number of days before or after individuals' 18th birthdays. Arrest counts are constructed as described in Section 3. The mean crime frequency reported in Table 2 reflects the average number of offenses committed in the pre-age of majority period. For violent crimes this is a mean of

[6] A surge could be induced by either false age reports, or offenders that wish to take advantage of juvenile sentencing before reaching the age of majority.

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208, and for property crimes this is a mean of 363. The running variable is individuals' ages in exact days from the age of 16 to 19 inclusive. The coefficient of interest is the over 18 indicator which reveals the deterrence effect of adult sanctions relative to juvenile sanctions on adolescents at the age of 18.

Each column in Table 2 represents a different empirical specification. The dependent variable in Columns (1) - (4) is the arrest count for violent crimes. The dependent variable in Columns (5) - (8) is the arrest count for property crimes. Panel A presents coefficient results from global parametric estimation. Columns (1), (2), (5) and (6) present results using a linear effect in age. Columns (3) and (7) present results using a second order global polynomial. Columns (4) and (8) present global third order parametric results. In all cases, the polynomial is allowed to differ on each side of the threshold. Dummy variables that control for birth date effects are included in Columns (2)-(4) and (6)-(8).

Panel B presents results from local nonparametric estimates using the triangular kernel and data-driven MSE-optimal bandwidth selection methods. Columns (1) and (4) report conventional RD estimates. Columns (2), (3), (5) and (6) report estimates from robust nonparametric bias-correction methods following Calonico, Cattaneo, and Titiunik (2014) and Calonico, Cattaneo, and Farrell (2017). Conventional standard errors are reported in Columns (1), (2), (4) and (5), while robust standard errors are reported in Columns (3) and (6).

Entries in the first rows of each panel are estimated regression discontinuities at age 18, as measured by the decline in the total daily crime count. Mean crime frequency is the baseline total average daily crime count committed by youths just below age 18. In panel A, where we use global parametric RD estimation, the pre-period mean is constructed using offenses for 2 years prior to the age discontinuity. In panel B, where we use local nonparametric estimation, we construct the pre-period mean under the MSE-optimal bandwidth. Entries in the last rows of both panel A and panel B are estimated regression discontinuities at age 18 measured as a percentage change in crime. This is calculated as the discontinuous drop in crime counts at age 18 divided by the baseline mean daily crime counts committed by youth just below age 18. This allows us to interpret the coefficient estimates as the change in crime relative to what crime would have been in the absence of an increase in criminal

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sanctions.

Across specifications and estimation techniques violent crimes drop by a considerable magnitude ranging from roughly 10% to 12%. So long as the previously discussed assumptions are satisfied, these estimates are measures of the deterrence effect. Estimates are all statistically significant at 1% across all specifications, and quite uniform in magnitude. These results supply strong evidence that adolescents are rational from the perspective that they meaningfully modify their criminal behavior when they are exposed to a harsher correctional regime. Figure 1 graphically depicts these results by showing precise daily age on the x-axis and pooled crime counts by precise age on the y-axis. We estimate and superimpose global polynomial estimates on the crime counts in Figure 1. The gap between the two polynomials at the age of 18 reveals the drop in crimes reported in Table 2.

Figure 2 reveals how the estimate of the treatment effect varies under the use of different bandwidths. We consider a wide range of bandwidths that vary from approximately 2 months to 3 years while estimating the regression discontinuity using the second order global polynomial approach. Figure 2 reveals that the point estimate is stable and in all cases the 95% confidence interval is always far from zero. As such, the impact of greater sanction intensity at the age of majority on violent crimes is in no way sensitive to the bandwidth used in estimating the RD coefficients.

Property crime estimates are generally smaller and less consistent across estimation techniques and specifications. Local nonparametric estimates, as well as third order global parametric estimates are all statistically significant at the 1% level and are highly supportive of a deterrence effect. Moreover, magnitudes are meaningfully large suggesting property crime declines by 8-9% as a result of the greater sanctions at the age of majority. However, first and second order global parametric estimates show much smaller magnitudes and lower levels of significance. The fact that these results are dependent upon functional form and model choice suggests that we need to interpret these results with a greater degree of caution. We view our results as suggestive evidence for a deterrence effect. Furthermore, it is important to note that the effect on property crimes are less likely to be cleanly identified as there exists the possibility of differential police behavior at the discontinuity that is not present in the case of violent crime. This issue is explored further in Section 5.

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5 Potential Identification Concerns

In a broad sense, there is a concern that police may treat offenses differently for juveniles and adults.[7] In some instances it is plausible that police may exercise some discretion in the choice to make an arrest or not. For example, one could imagine a context where police may view a drug possession arrest as having lasting deleterious effects on youths' future, as noted in Thompson (2016), and opt not to arrest. Additionally, law enforcement officials may feel less inclined to arrest a juvenile if they perceive that the arrest is unlikely to result in further punitive action as noted by Goldman (1963). In contrast, an adult would likely not be the beneficiary of such police benevolence. It is also arguably the case that police officers may respond to the additional burdens of arresting juveniles, such as parental notification and greater documentation, by decreasing their propensity to arrest in marginal cases. All of these possibilities would serve to make detection of a deterrence effect far more difficult than in their absence. It is also conceivable that law enforcement may be harsher with juveniles. Although this is a possibility, there has been a literature supporting the assertion the police are considerably more likely to arrest adults than juveniles rather than the opposite.[8]

Differential police behavior is most likely to be present for petty, non-violent, or victimless crimes. To ensure that our primary analysis does not suffer from the potential problems of differential police behavior, we restrict the sample to violent crimes. Violent offenses are highly unlikely to suffer from this problem as the very nature of the crime is serious and creates victims. It is quite implausible that police would choose to turn a blind eye to serious violent offenses such as murder, robbery, rape, and assault. Hence, our main analysis focuses on these crimes, as here identification is clean from the perspective of differential police behavior. An additional motivation for considering violent crimes is that, as noted by Chalfin and McCrary (2017), these crimes impose the greatest social costs with an average cost of $1,798,392 per violent crime.

Another potential concern is that it would be problematic for our identification strategy

[7] One obvious case where this would occur would be status offenses, which are crimes, but only for minors. A few crimes, such as statutory rape, are also asymmetrically defined. However, these asymmetric offenses are crimes for adults, but non-offenses for juveniles. In our analysis we exclude all status offenses and asymmetrically defined crimes to ensure appropriate comparisons are made.

[8] See, for example, Liederbach (2007), Sun and Payne (2004), and Goldman (1963).

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if it were the case that considerable numbers of juveniles are passed on to adult courts to face adult penalties. If juveniles have a reasonable expectation of facing adult penalties prior to the age of majority then this fact alone could alter criminal behavior prior to age 18 and potentially mask the presence of any treatment effect at the discontinuity. However, we view this as unlikely to be the case given the near extreme rarity with which juveniles are transferred to adult courts in California. As noted by the California Bureau of Criminal Information and Analysis about 0.02% of arrests result in charges being brought in adult criminal court within California during our sample period. Thus, transfer to adult court is exceedingly rare and insufficient to bias the RD estimates in any meaningful way.

Although penalties can deter crime, penalties also have the ability to alter crime through a second channel known as the incapacitation effect. As such, we believe a discussion of deterrence effects versus incapacitation effects is merited. Once an offender is apprehended, if penalties are sufficiently harsh, that offender will be removed from circumstances that allow them to commit further crimes against the public.[9] Incapacitation effects are not a concern with RD estimates as they would not result in any discontinuous changes in criminal acts at the date of majority. Incapacitation effects, in this context, are caused by the increased duration of adult incarceration. These effects start to emerge after individuals are retained in the adult system for a time period longer than they would have been under the juvenile system. This results in a continuous trend with a smooth decline in the slope soon after the date they turn 18. Loeffler and Chalfin (2017) report that incapacitation effects arise roughly 6 months after the age of majority, while Lee and McCrary (2017) argue that incapacitation effects could potentially emerge as early as 30 days after the age of majority. In either case, incapacitation effects do not lead to a discontinuity in trends at, or immediately after, the date adolescents turn 18. Thus, within our RD framework, incapacitation effects do not obscure the deterrence effect.

Another concern in our context would be the effect of *en masse* dropping out of school, since 18 is the compulsory schooling age in California. Any abrupt change in the fraction of adolescents in school at 18 could lead to differences in criminal propensity if schools provide incapacitation effects as noted by Jacob and Lefgren (2003), and Carr and Doleac (2016).

[9] In many circumstances the incapacitation effect is referred to as an incarceration effect.

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We use ACS and Decennial Census data to investigate whether or not there are smooth trends in school enrollment. Figure 3 shows that although school attendance declines in the late teens, the trend is smooth and the slope does not change at age 18. Other potential changes at 18 that could potentially alter criminal acts are employment, independent living as opposed to living with parents, and wage earnings.[10] Weiss (1988) notes that there is evidence for a wage premium associated with completion of high school, but that there is no evidence for a wage premium at age 18. Given that high school completion does not coincide with precise passage to the age of majority, we argue that any high school completion wage change does not pose a threat to identification. Figure 3 also shows trends for these variables using ACS and Decennial Census data. Again, we find strong evidence of smooth trends throughout the late teens and early twenties.

6 Extensions and Additional Evidence

6.1 Heterogeneity in Deterrence

While our main findings suggest strongly that deterrence effects exist and are fairly large from a practical perspective, we explore the degree to which the magnitudes of deterrence effects are heterogenous across demographic groups. While explaining the causal reasons for heterogenous effects is beyond the scope of this paper, we find the heterogeneity of effects noteworthy and worth careful consideration.[11] We offer plausible explanations for heterogeneity where appropriate.

Race

We partition our sample into observations from particular racial categories and re-estimate deterrence effects. In particular, we estimate effects for white, Hispanic, black and Asian offenders. Results are shown in Table 3. Coefficients report the change in the daily mean crime level at the age discontinuity relative to the daily mean crime level immediately before

[10] California youth may obtain a driver's license at 16. Hence, there is no discontinuous change in access to transportation at age 18.

[11] In unreported results, that are available upon request, we find that effects are not being driven by a particular subset of counties and that effects are similar across counties.

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the discontinuity. To interpret coefficients as percentage changes we compare the point estimates to the mean crime level prior to adolescents reaching the age of majority. Figure 4 visually depicts these results by showing crime counts by precise daily age, superimposing fitted global second order polynomials on the crime counts.

For white, Asian, and Hispanic race groups effects are highly significant from a statistical perspective and all four race categories present estimates with the expected negative sign. In the case of black youth, although the coefficient is negative, the effect is indistinguishable from zero. As in our main analysis, our results remain highly consistent under both the global parametric and the local nonparametric frameworks. Reassuringly, these results suggest that our results are not being driven by any one particular demographic group. Furthermore, racial differences in deterrence offer additional insights that may be of considerable interest to both law enforcement officials and policy makers.

Effects are largest for white and Asian adolescents. White adolescent crime decreases by about 19%, and Asian crimes decline by about 16% as a response to the greater punitiveness of adult sanctions. This would suggest that violent crimes committed by white and Asian offenders are to some degree preventable via greater sanctions. Given that white adolescents carry out 29.4% of all violent crime committed by adolescents, the reduction in violent crimes via white offenders translates to a 5.6% decline in the total number of violent offenses carried out by adolescents. This is a practically significant reduction, especially in light of the high costs associated with violent crime.

Hispanic adolescents appear to alter criminal behaviors to a lesser, but still significant, degree. Hispanic violent crimes fall by about 7% as a result of criminal deterrence. This suggests that violent crimes committed by Hispanic adolescents are also potentially preventable, but are less responsive to increased sanctions than criminal acts carried out by white and Asian offenders. However, within California 43.4% of youth committed violent offenses are carried out by Hispanic individuals. As such the 7% decline in violent offenses is quite meaningful and reduces the total prevalence of youth violent crimes by a very considerable 3%.

Our results suggest that black adolescents may represent an outlier in that they appear somewhat non-responsive to the change in incentives. One explanation would simply be that

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elasticities of crime with respect to sanctions fundamentally vary by race. If this is the case, our results would suggest that black youth have smaller elasticities than Asian and white youth. An alternative potential explanation would be that black youth may face a relatively more punitive juvenile system where punishments are more similar to those experienced by adult offenders. Similarly, white, Hispanic, and Asian youth may face a juvenile system that is qualitatively less harsh than what black youth face. Eren and Mocan (2018) provide compelling evidence that black youth may indeed face harsher sanctions within the juvenile system that juveniles of other races. If this is indeed true, black youth would face a smaller increase in relative punitiveness at the age of majority and would rationally alter criminal behavior to a lesser degree at age 18. Another explanation would be that perhaps black adolescents are less aware of the large change in sanctions at the age of majority. Finally, it is possible that post age 18 economic opportunities are systematically worse for black youth. The possibility that black youth face an inherently lower level returns to legal work may alter behavioral responses to changes in sanction intensity. This explanation is potentially consistent with Grogger (1998), which notes racial wage differentials are meaningful in the context of crime.

Gender

In addition to racial differences, there may be meaningful differences between gender groups. We estimate deterrence effects by gender and present results in Table 4. Both male and female adolescents display behaviors consistent with sizable deterrence effects. As in prior results, our gender based results are consistent across both the global parametric and local linear frameworks. Figure 5 visually depicts these results by superimposing fitted global second order polynomials on the crime counts by age.

Our results suggest female adolescents are relatively more responsive as violent criminal acts drop by about 17% at the age of majority as a result of deterrence. For males the response is still large and meaningful, but relatively smaller. Male violent crime falls by about 8%. This is consistent with findings by Kling, Ludwig, and Katz (2005) that note greater responses to changes in incentives and neighborhoods by female offenders. These findings suggest that male and female adolescents have different elasticities of criminality

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with respect to punishment severity. In this case, females may be more elastic than their male counterparts.

6.2 Nonparametric Event Study

We employ a nonparametric event study as seen in Jacobson, LaLonde, and Sullivan (1993) and more recently in Dobkin et al. (2018), to investigate the frequency of violent crime relative to the event of reaching the age of majority. The primary advantage of utilizing a nonparametric event study is that it allows us to visually assess the pattern of outcomes relative to the month where youth reach the age of majority. Thus it is possible to observe the time-varying evolution of the effect of a more stringent justice system on adolescent criminal behavior. It should be noted that the RD identification strategy is generally more reliable since the RD relies on weaker identifying assumptions. However, the event study reveals the time trend of frequency of crime over the course of a few years after reaching the age of majority, which is an aspect we are not able to assess under an RD strategy, which presents reliable, but very local effects at precisely the age of majority. Additionally, the event study allows for incorporation of county and time fixed effects. In the case that there were a time, or county, specific component of the error term that is correlated with the timing of treatment, fixed effects would address this problem.

It should be noted that as we leave the RD specification and move to an event study framework, we are less able to cleanly separate the deterrence effect from the incapacitation effect. As individuals move further from the precise age cut-off there exists the possibility that as time moves forward, more criminals will be rendered unable to commit crime by virtue of being incapacitated via the greater sanctions of the adult justice system. Moreover, life cycle effects must also be considered as we move away from the discontinuity at age 18 as people age in and out of criminal acts.[12]

The primary coefficients of interest under the nonparametric event study framework are the violent crime frequencies over time relative to the event of reaching the age of majority.

[12]Criminal behaviors may change over the life cycle as offense patterns change with age. In particular, criminal acts are likely to increase with age in the months and years around the age of majority. This is particularly true for violent offenses which typically do not reach their peak until the early twenties.

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We define age relative to the age of majority as event time. The event time is distinct from calendar time, which is the calendar year and month in which the crime occurred. The nonparametric event study specification takes the form:

$$Crime_{it} = X_{it}\alpha + \sum_{r=S}^{r=-2} \beta_r + \sum_{r=0}^{r=F} \beta_r + \gamma_t + \varepsilon_{it} \tag{4}$$

where β_r are coefficients on event time indicators for month relative to the month where an individual reaches the age of majority. This allows us to visually investigate the pattern of crime trends in months and years after turning 18. X_{it} represents a vector of other potential control variables, γ_t are coefficients on calendar time fixed effects. All analyses allow for an arbitrary variance-covariance matrix. S and F are the start period and the end period in our analysis. In our event study analysis we focus on 16 to 20 year old youths, as such $S = -24$ and $F = 35$.

The identifying assumption behind the event study is that the timing of the treatment at 18 is uncorrelated with the outcome. One way this assumption would be violated is if the error term is systematically correlated with the timing of passage to the age of majority. However, our prior results using ACS and Decennial Census data suggest this is unlikely.

Another way the identifying assumption would be violated is if there are time-varying shocks that are correlated with both the timing of passage to the age of majority and criminal propensity. In this case, this is the stronger of the two assumptions. An important consideration is that both the life-cycle effect and incapacitation effects, especially due to greater stringency at age 18, would become increasingly likely to violate the identifying assumption as event time progresses further from the age of majority. As a result, we caution that estimates should not be interpreted as the causal impact of increased sanctions over time, but rather reveal the nature of how crime trends evolve relative to the age of majority. To what extent those trends are driven by deterrence effects, life-cycle effects, or incapacitation effects is less than perfectly clear.

Figure 6 shows results from the nonparametric event study. The x-axis reveals age relative to the age of majority, while the y-axis reports the incidence of violent crime relative to violent crime in the baseline month, which is the last month before individuals reach the

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age of majority. Each unit of observation is a violent crime count, committed by a precise age group in months, in a calendar month and year in a particular county. The mean level of violent crime in the baseline period is 1.25. We control for year and month fixed effects, as well as county fixed effects. Confidence intervals are shown as dashed lines and rely on standard errors clustered at the county level. Violent crimes drop from 1.35 to 1.05 in the month that individuals are exposed to the greater punitiveness of the adult criminal justice system. This implies an effect of about 12% and is quite consistent with out RD estimates that varied within a 10% to 12% reduction window. It merits noting that at this point in event time, incapacitation effects and life-cycle effects have not had time to introduce meaningful bias into the estimates. Hence, this 12% decline represents a deterrence effect, as in the RD estimates. As we move away from the age threshold, it should be noted that violent crime typically peaks in the early twenties, as such life-cycle effects should attenuate our estimates.

The time-varying evolution of violent crime displays a considerable decline in the aftermath of passage to the age of majority, and then rises slightly while still remaining well below the level of offending prior to the age of majority. These trends must be interpreted with some degree of caution as age and incapacitation effects will introduce bias as we move further away from the 18th birthday. One interpretation is that in the absence of exposure to greater sanctions, violent crimes would have continued at a similar level as the fairly stable pre-period level of violent offenses. This would suggest that the age of majority induces a fairly permanent and stable reduction in violent crime, but that age or life-cycle effects may mask deterrence effects as the passage of time continues.

6.3 Elasticity

Our findings imply that adolescents in California do respond to greater sanctions, and in a fairly meaningful way. However, policy makers that have some ability to alter the sanctions regime would presumably be equally interested in how the deterrence magnitude compares to the magnitude of the change in sanctions. In this context, an elasticity measure should prove useful and enlightening. It should be noted that the data requirements to generate an optimal elasticity measure are likely very high owing to both qualitative and quantitative

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considerations in measuring sanctions that stem from the need to accurately capture the relative severity of the adult justice system versus the juvenile justice system.

Qualitatively, there is ample evidence to suggest that juvenile detention facilities are considerably different than adult facilities with a much greater emphasis on education and reform. Moreover, there is evidence that within crime categories juvenile offenses are less egregious. Finally, there is an argument to be made that youth may feel higher psychic costs associated with incarceration than adults. Thus, regardless of data on sentence duration, comparisons between adult systems and juvenile systems are difficult endeavors that present data is not entirely able to capture. Thus our elasticity measure, while important, is necessarily an imperfect measure.

Quantitatively, expected sentence duration conditional on committing a crime would be the measure of choice for quantifying the punitiveness of a sanction regime. Unfortunately, expectations are unobservable, and California juvenile sentencing duration data is not available. Given these data limitations, we adopt the approach of Levitt (1998) and consider the ratio of incarcerated offenders to arrested violent offenders for both juveniles and adults. Arrest data is obtained via the MACR, adult incarceration data is from the Census of State and Federal Adult Correctional Facilities, while juvenile incarceration data is from the Census of Juveniles in Residential Placement. It should be noted that the measure clearly captures the probability of incarceration conditional on arrest (the extensive margin), but since longer sentences mean a higher incarcerated population, it also partially captures the sentence duration (the intensive margin) for both juveniles and adults.

Using these measures, we conclude that the adult justice system is roughly 69% more punitive than the juvenile system. In conjunction with our estimates of a 10-12% reduction in violent crimes as a result of the greater sanctions this yields a crime with respect to sanction intensity elasticity that ranges from -0.145 to -0.174. This would place our estimates a higher than Lee and McCrary (2017), but close to their upper bound, and less than Drago, Galbiati, and Vertova (2009). The former relies on a structural approach using data on Florida adolescents, while the population of interest in the latter study consists of repeat, adult offenders that have near perfect information about the increase in criminal sanctions stemming from a third offense. We posit that these differences may explain some of the

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variance in elasticity estimates.

6.4 Reconciliation with Prior Results

A natural question that merits further consideration is why our study has found evidence of deterrence for youthful offenders, while other studies have uncovered more limited evidence. While a precise and irrefutable answer is beyond the scope of this study, there are several considerations that may help to reconcile our results with prior studies, particularly Lee and McCrary (2017) which is methodologically similar, but leverages data from the state of Florida.

Perhaps the most readily apparent reason for differing patterns of effects is the considerable heterogeneity in states' respective juvenile and adult corrections systems. The key mechanism for the creation of deterrence effects at the age of majority is the sharp change in incentives that youth face, which in turn depends upon the magnitude of the jump in sanction intensity between juvenile and adult justice systems. The larger the jump in sanctions, the stronger the incentive to desist from offending; hence, the ideal institutional setting for uncovering deterrence effects would be a lenient juvenile system paired with a harsh adult system.

In California, it is plausible that the jump in sanctions between the juvenile and adult justice systems may be substantially larger than in many other states. With respect to the adult corrections system, California was notorious for abysmal adult prison conditions and overcrowding during the years that we observe. Prison conditions and overcrowding were sufficiently appalling that the U.S. Supreme Court's 2011 ruling in the *Brown v. Plata* case mandated California to substantially reduce the number of incarcerated offenders in order to return prison conditions to an acceptable level. Consistent with the findings of Katz, Levitt, and Shustorovich (2003), the presence of reprehensible prison conditions likely had deterrence value. On the other side of the age threshold, California is generally not held up as a paragon of harsh juvenile justice, as many Southern states have been. In particular, California virtually never transfers juvenile offenders to adult court, while Florida leads the nation in this regard and detains more juveniles in harsher adult facilities than the lowest

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28 states combined.[13] Hence, it is inherently plausible that along qualitative dimensions California represents a state where the gap in sanction intensity is large, and therefore conducive to eliciting a deterrent effect at the age of majority.

A second reason California may be predisposed to greater deterrence effects is demographic. Our findings provide evidence that deterrence effects are greatest for white and Asian youth, while being largely absent for black youth. Florida and California are quite different in demographic ways that may lead to greater deterrence in California. Notably, California has the second highest Asian population share in the nation, and for a large state, a relatively small black population share. Florida's demographics are more similar to rest of the Southeastern US with a small Asian population share, and a fairly large black population share. Given that we find large deterrence effects for Asian youth, and little evidence for deterrence among black youth. It is quite plausible that demographic differences help reconcile our findings with the more modest deterrence effects found by Lee and McCrary (2017). Indeed, re-weighting our central estimates for violent crime using population weights from Florida demographics rather than California, yields a 25% reduction in deterrence effects.

A final consideration is that our findings are uncovered using the universe of all arrests across 9 years in what is by far the largest US state, with a population of roughly 39 million. While use of population data from a very populous state does not drive deterrence, the tremendous number of observations likely makes detection of deterrence effects a more manageable task than in less data-rich settings.

In a more general sense, uncovering deterrence effects at the age of majority is likely a challenging exercise and may explain prior negative results in the literature. As discussed by Arora (2018), there are several considerations at play that may mask or attenuate deterrence effects at the age of majority. Notably, Arora (2018) uses incident data from New England states to document that in states where the age of majority is 17, there is a relative dearth of reported incidents for juveniles prior to the age of 17; however, no such dearth of incidents exists in nearby states where the age of majority is 18. Similarly, in states where the age of majority is 18, there is a relative dearth of official incidents prior to age 18, but not in states where the age of majority is 17. This pattern suggests the detection of deterrence effects is

[13] Source: Office of Juvenile Justice and Delinquency Prevention.

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indeed a difficult task. Provided this reporting behavior exists, it is likely due either to a desire to treat juvenile offenders more generously, or due to additional practical constraints that surround juvenile arrests and may serve to de-motivate police officers. For example, juveniles may only be detained for a maximum of 48 hours while charges are being filed, juveniles cannot be detained in county jails, parents and guardians must be notified of their child's arrest, and more generally juvenile arrests create a higher documentation burden for law enforcement.[14]

A final reason that deterrence effects may be difficult to uncover is that the age of majority occurs at a time in youth's lives when the age profile of offending is continuing to increase. That is, in the absence of changes in the sanction regime, adolescent crimes would be expected to rise in a secular manner as youth grow older. Thus, the age profile of offending may mask the presence of deterrence effects. The presence of age-profile effects is likely of importance for studies leveraging difference-in-differences empirical strategies, where the post period is sufficiently long that life-cycle effects may render deterrence effects difficult to uncover.

6.5 Policy Considerations

In recent years Connecticut, Illinois, Louisiana, Massachusetts, Mississippi, Missouri, New Hampshire, New York, North Carolina and South Carolina have all passed "raise the age" reforms raising the age of majority to 18 years. Contemporaneously, other states such as Texas and Wisconsin have considered, but as of yet, failed to enact such legislation. Additionally, there have been legislative efforts to further raise the age of majority to 21 in Illinois and Connecticut, while Vermont has successfully passed such legislation. Proponents of these reforms have argued that youthful offenders lack the maturity to properly process the threat of harsher sanctions (Steinberg and Scott (2003)). Thus, exposing youthful offenders to harsher sanctions fails to deter crime, but instead imposes additional costs to offenders in the form of lost earnings, taxpayers in the form of higher incarceration costs, and future victims in the form of greater recidivism stemming from acquisition of criminogenic capital,

[14]Source: California Attorney General's office.

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and the "brutalizing effects" of adult prisons.[15] While these putative cost savings are likely legitimate, it is important to recognize that our study implies the presence of substantial social savings that result from deterrence of crimes via greater sanctions imposed at the age of majority. Moreover, there are likely some cost savings that stem from the fact that juvenile incarceration costs far outpace the costs of incarceration in adult facilities.[16] In light of our findings, it would be disingenuous to advocate for raise the age policies as costless. Rather, the loss of deterrence effects would need to be weighed against the putative benefits of an increased age of majority.[17]

7 Conclusion

Although economic theory has long and unambiguously posited that greater severity of punishments in the criminal justice system will reduce crime through a deterrence channel, clear and consistent empirical evidence has proven somewhat elusive. Many studies find little evidence of a deterrence effect, while others find evidence of more sizable effects. We address this issue by using administrative data from California that covers the universe of adult and juvenile crime and allows us to estimate the deterrent effect of increased criminal sanctions that occur when adolescents reach the age of majority. Employing a regression discontinuity design, under a wide range of specifications and modeling choices, our results point to clear, significant, and large deterrence effects for violent crime. Estimates suggest that the greater degree of punitiveness imposed upon adolescents at the age of majority deters violent crimes by 10-12%. We also consider property crimes and find a reduction of up to 8% that must be interpreted with caution as there exists the possibility of differential police behavior at the age cut-off.

We also extend our results by considering racial and gender populations that suggest deterrence effects are potentially quite heterogenous. Female offenders are generally more

[15] Whether juvenile detention increases recidivism remains an issue of much interest, but with mixed findings. While Aizer and Doyle Jr (2015) find evidence that detention increases the likelihood of recidivism, Loeffler and Grunwald (2015a) find stiffer punishments at the age of majority may not lead to changes in recidivism.

[16] Source: Justice Policy Institute (2014).

[17] Moreover, there may also be philosophical considerations as argued by Yaffe (2018) that merit careful consideration as well.

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responsive than their male counterparts. Similarly, white and Asian offenders appear quite responsive while Hispanic offenders are slightly less responsive. Interestingly, black adolescents appear generally non-responsive to the changes in the severity of criminal sanctions. Explaining this feature of the data would appear to be a fruitful avenue for future research. Potentially, there may be differences in the extent to which adolescents are aware of the change in criminal sanctions at the age of majority. If this is true, it would appear a highly cost effective policy would be aimed at increasing adolescent awareness of the change in criminal sanctions.

Finally, to contextualize our results we estimate the rough magnitude of increased sanctions in the adult justice system and offer an elasticity measure that sheds light on how responsive adolescents are relative to the increase in sanctions. Based on a violent crime reduction of 10-12%, our elasticity measure varies from -0.145 and -0.174.

Throughout the 1990s and 2000s incarceration rates have unambiguously risen despite declines in both violent and property crimes. It increasingly appears that we as a society are now at a cross-roads where some are in favor of reduced sanctions, while others are not. Our findings should be of considerable interest in this debate. However, a complete understanding of the issue must bear in mind that incarceration may make individuals more, or less likely to commit future crimes. Experiencing incarceration may be a deterrent, or it may harden offenders and afford them opportunities to gain criminogenic human capital. Without fully understanding these matters, optimal policy will likely remain elusive.

We also speculate that our findings may offer some insight into productive avenues of future research. One problem with the age of majority literature is that although it is non-controversial that sanctions increase, this occurs through two channels. Both the certainty and severity of punishment increase, but we are unable to separate their effects. Additionally, we find strong evidence of heterogenous deterrence effects. While we can offer plausible explanations, it would be of great value to determine what drives these heterogenous responses. Finally, better measures of the increase in the punitiveness of sanctions at the age of majority would allow for more credible elasticity calculations. Further work in all of these areas would presumably be highly valuable for policy makers that must operate in a high stakes environment.

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References

Aizer, Anna and Joseph J Doyle Jr. 2015. "Juvenile Incarceration, Human Capital, and Future Crime: Evidence from Randomly Assigned Judges." *The Quarterly Journal of Economics* 130 (2):759–803.

Angrist, Joshua D and Alan B Keueger. 1991. "Does Compulsory School Attendance Affect Schooling and Earnings?" *Quarterly Journal of Economics* 106 (4):979–1014.

Arora, Ashna. 2018. "Juvenile Crime and Anticipated Punishment."

Beccaria, Cesare. 1764. *On Crimes and Punishments*. Oxford: Clarendon Press.

Becker, Gary S. 1968. "Crime and Punishment: An Economic Approach." *The Journal of Political Economy* 76 (2):169–217.

Bentham, Jeremy. 1789. *An Introduction to the Principles of Morals and Legislation*. Oxford: Clarendon Press.

Blumstein, Alfred, Jacqueline Cohen, Jeffrey A. Roth, and Christy A. Visher. 1986. *Criminal Careers and "Career Criminals"; Volume II*. Washington, DC: The National Academies Press.

Calonico, Sebastian, Matias D. Cattaneo, and Max H. Farrell. 2017. "On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference." *Journal of American Statistical Association* .

———. 2018. "Coverage Error Optimal Confidence Intervals."

Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik. 2014. "Robust Nonparametric Confidence Intervals for Regression-discontinuity Designs." *Econometrica* 82 (6):2295–2326.

Carpenter, Christopher and Carlos Dobkin. 2015. "The Minimum Legal Drinking Age and Crime." *Review of Economics and Statistics* 97 (2):521–524.

Electronic copy available at: https://ssrn.com/abstract=3116414

Carr, Jillian B and Jennifer L Doleac. 2016. "Keep the Kids Inside? Juvenile Curfews and Urban Gun Violence." *Working Paper* .

Chalfin, Aaron and Justin McCrary. 2017. "Criminal Deterrence: A Review of the Literature." *Journal of Economic Literature* 55 (1):5–48.

Cheng, Ming-Yen. 1997. "Boundary Aware Estimators of Integrated Density Derivative Products." *Journal of the Royal Statistical Society* 59 (1):191–203.

Cook, Philip J and Songman Kang. 2016. "Birthdays, Schooling, and Crime: Regression-discontinuity Analysis of School Performance, Delinquency, Dropout, and Crime Initiation." *American Economic Journal: Applied Economics* 8 (1):33–57.

Di Tella, Rafael and Ernesto Schargrodsky. 2004. "Do Police Reduce Crime? Estimates Using the Allocation of Police Forces After a Terrorist Attack." *American Economic Review* 94 (1):115.

Dobkin, Carlos, Amy Finkelstein, Raymond Kluender, and Matthew J Notowidigdo. 2018. "The Economic Consequences of Hospital Admissions." *American Economic Review* 108 (2):308–52.

Donohue, John J and Justin Wolfers. 2005. "Uses and Abuses of Empirical Evidence in the Death Penalty Debate." *Stanford Law Review* 58 (3):791.

Draca, Mirko, Stephen Machin, and Robert Witt. 2011. "Panic on the streets of london: Police, crime, and the july 2005 terror attacks." *American Economic Review* 101 (5):2157–2181.

Drago, Francesco, Roberto Galbiati, and Pietro Vertova. 2009. "The Deterrent Effects of Prison: Evidence from a Natural Experiment." *Journal of Political Economy* 117 (2):257–280.

Ehrlich, Isaac. 1973. "Participation in Illegitimate Activities: A Theoretical and Empirical Investigation." *Journal of Political Economy* 81 (3):521–565.

Electronic copy available at: https://ssrn.com/abstract=3116414

Eren, Ozkan and Naci Mocan. 2018. "Emotional Judges and Unlucky Juveniles." *American Economic Journal: Applied Economics* 10 (3):171–205.

Gelman, Andrew and Guido Imbens. 2014. "Why High-order Polynomials Should Not Be Used in Regression Discontinuity Designs." National Bureau of Economic Research Working Paper No. 20405.

Glassner, Barry, Margret Ksander, Bruce Berg, and Bruce D Johnson. 1983. "A Note on the Deterrent Effect of Juvenile vs. Adult Jurisdiction." *Social Problems* 31 (2):219–221.

Goldman, Nathan. 1963. *The Differential Selection of Juvenile Offenders for Court Appearance*. National Research and Information Center, National Council on Crime and Delinquency.

Grogger, Jeff. 1998. "Market Wages and Youth Crime." *Journal of Labor Economics* 16 (4):756–791.

Hahn, Jinyong, Petra Todd, and Wilbert Van der Klaauw. 2001. "Identification and Estimation of Treatment Effects with a Regression-discontinuity Design." *Econometrica* 69 (1):201–209.

Hansen, Benjamin. 2015. "Punishment and Deterrence: Evidence from Drunk Driving." *American Economic Review* 105 (4):1581–1617.

Hekman, Randall J, Martin Gold, and David J Ruhland. 1983. "Reducing Crime: It Can Be Done." *Juvenile and Family Court Journal* 34 (3):3–7.

Helland, Eric and Alexander Tabarrok. 2007. "Does Three Strikes Deter? A Nonparametric Estimation." *Journal of Human Resources* 42 (2):309–330.

Hjalmarsson, Randi. 2008. "Crime and Expected Punishment: Changes in Perceptions at the Age of Criminal Majority." *American Law and Economics Review* 11 (1):209–248.

———. 2009. "Juvenile Jails: A Path to the Straight and Narrow or to Hardened Criminality?" *Journal of Law and Economics* 52 (4):779–809.

Electronic copy available at: https://ssrn.com/abstract=3116414

Imbens, Guido and Karthik Kalyanaraman. 2012. "Optimal Bandwidth Choice for the Regression Discontinuity Estimator." *Review of Economic Studies* 79 (3):933–959.

Iyengar, Radha. 2008. "I Would Rather Be Hanged for a Sheep than a Lamb: The Unintended Consequences of 'Three-strikes' Laws." National Bureau of Economic Research Working Paper No. w13784.

Jacob, Brian A and Lars Lefgren. 2003. "Are Idle Hands the Devil's Workshop? Incapacitation, Concentration, and Juvenile Crime." *American Economic Review* 93 (5):1560–1577.

Jacobson, Louis S, Robert J LaLonde, and Daniel G Sullivan. 1993. "Earnings Losses of Displaced Workers." *American Economic Review* :685–709.

Justice Policy Institute. 2014. "Sticker Shock: Calculating the Full Price Tag for Youth Incarceration."

Katz, Lawrence, Steven D Levitt, and Ellen Shustorovich. 2003. "Prison Conditions, Capital Punishment, and Deterrence." *American Law and Economics Review* 5 (2):318–343.

Kling, Jeffrey R. 2006. "Incarceration Length, Employment, and Earnings." *American Economic Review* 96 (3):863–876.

Kling, Jeffrey R, Jens Ludwig, and Lawrence F Katz. 2005. "Neighborhood Effects on Crime for Female and Male Youth: Evidence from a Randomized Housing Voucher Experiment." *Quarterly Journal of Economics* 120 (1):87–130.

Krisberg, Barry, Christopher Hartney, and Susan Marchionna. 2010. "A New Era in California Juvenile Justice: Downsizing the State's Youth Corrections System." .

Lee, David S and Justin McCrary. 2017. "The Deterrence Effect of Prison: Dynamic Theory and Evidence." *Advances in Econometrics* .

Levitt, Steven D. 1997. "Using Electoral Cycles in Police Hiring to Estimate the Effect of Police on Crime." *American Economic Review* :270–290.

———. 1998. "Juvenile Crime and Punishment." *Journal of Political Economy* 106 (6):1156–1185.

Electronic copy available at: https://ssrn.com/abstract=3116414

Levitt, Steven D and Lance Lochner. 2001. "The Determinants of Juvenile Crime." In *Risky behavior among youths: An economic analysis*. University of Chicago Press, 327–374.

Liederbach, John. 2007. "Controlling Suburban and Small-town Hoods: An examination of Police Encounters with Juveniles." *Youth Violence and Juvenile Justice* 5 (2):107–124.

Lochner, Lance. 2007. "Individual Perceptions of the Criminal Justice System." *American Economic Review* 97 (1):444–460.

Lochner, Lance and Enrico Moretti. 2004. "The Effect of Education on Crime: Evidence from Prison Inmates, Arrests, and Self-Reports." *American Economic Review* :155–189.

Loeffler, Charles E and Aaron Chalfin. 2017. "Estimating the Crime Effects of Raising the Age of Majority." *Criminology & Public Policy* 16 (1):45–71.

Loeffler, Charles E and Ben Grunwald. 2015a. "Decriminalizing Delinquency: The Effect of Raising the Age of Majority on Juvenile Recidivism." *The Journal of Legal Studies* 44 (2):361–388.

———. 2015b. "Processed as an Adult: A Regression Discontinuity Estimate of the Crime Effects of Charging Nontransfer Juveniles as Adults." *Journal of Research in Crime and Delinquency* 52 (6):890–922.

Machin, Stephen and Costas Meghir. 2004. "Crime and Economic Incentives." *Journal of Human Resources* 39 (4):958–979.

Marvell, Thomas B and Carlisle E Moody. 1996. "Specification Problems, Police Levels, and Crime Rates." *Criminology* 34 (4):609–646.

McCrary, Justin. 2008. "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test." *Journal of Econometrics* 142 (2):698–714.

Ruggles, Steven, J. Trent Alexander, Katie Genadek, Ronald Goeken, Matthew B. Schroeder, and Matthew Sobek. 2010. *Integrated Public Use Microdata Series: Version 5.0 [Machine-readable database]*. Minnesota Population Center [producer and distributor], Minneapolis, MN.

Electronic copy available at: https://ssrn.com/abstract=3116414

Steinberg, Laurence and Elizabeth S Scott. 2003. "Less Guilty by Reason of Adolescence: Developmental Immaturity, Diminished Responsibility, and the Juvenile Death Penalty." *American Psychologist* 58 (12):1009.

Sun, Ivan Y. and Brian K. Payne. 2004. "Racial Differences in Resolving Conflicts: A Comparison Between Black and White Police Officers." *Crime & Delinquency* 50 (4):516–541.

Thompson, Owen. 2016. "Drug Policy and Intergenerational Income Mobility in the United States." *Contemporary Economic Policy* 34 (1):127–145.

Waldfogel, Joel. 1994. "Does Conviction Have a Persistent Effect on Income and Employment?" *International Review of Law and Economics* 14 (1):103–119.

Weiss, Andrew. 1988. "High School Graduation, Performance, and Wages." *Journal of Political Economy* 96 (4):785–820.

Yaffe, Gideon. 2018. *The Age of Culpability: Children and the Nature of Criminal Responsibility*. Oxford University Press.

Electronic copy available at: https://ssrn.com/abstract=3116414

Table 1: Summary Statistics of Offenders

	Violent Adolescent Sample		Property Adolescent Sample	
Variable	Mean (1)	Standard Deviation (2)	Mean (3)	Standard Deviation (4)
White	0.267	0.442	0.322	0.467
Hispanic	0.442	0.497	0.427	0.495
Black	0.231	0.422	0.175	0.380
Asian	0.025	0.157	0.037	0.190
Male	0.818	0.386	0.755	0.430
Female	0.182	0.386	0.245	0.430
Age at Arrest	17.943	1.421	17.780	1.376
Small County Crimes	0.021	0.142	0.016	0.127
Medium County Crimes	0.139	0.346	0.121	0.327
Large County Crimes	0.841	0.366	0.862	0.345
Observations	358,664		574,444	

	Violent Full Sample		Property Full Sample	
Variable	Mean (1)	Standard Deviation (2)	Mean (3)	Standard Deviation (4)
White	0.330	0.470	0.345	0.475
Hispanic	0.394	0.489	0.397	0.489
Black	0.219	0.413	0.191	0.393
Asian	0.021	0.145	0.032	0.175
Male	0.805	0.396	0.723	0.447
Female	0.195	0.396	0.277	0.447
Age at Arrest	29.148	11.877	25.304	11.474
Small County Crimes	0.024	0.153	0.017	0.130
Medium County Crimes	0.133	0.340	0.117	0.322
Large County Crimes	0.843	0.364	0.866	0.341
Observations	1,929,670		1,986,903	

Notes: This table presents summary statistics for offenders arrested for violent crimes and property crimes. The top panel presents summary statistics for adolescent offenders between 16 and 20 years old. The bottom panel presents summary statistics for the full sample

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Table 2: Estimated Discontinuities in Violent and Property Crimes at Age 18

	Violent Crimes				Property Crimes			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Global Parametric Estimates								
RD at 18	-22.03***	-22.36***	-21.10***	-24.40***	-1.708	-1.513	-5.341*	-22.33***
	(1.569)	(1.563)	(2.255)	(3.000)	(2.232)	(2.234)	(3.086)	(4.023)
Age in days (Left)	-0.00636**	-0.00636**	-0.0885***	-0.119***	-0.0387***	-0.0387***	-0.177***	-0.168***
	(0.00263)	(0.00262)	(0.0101)	(0.0251)	(0.00375)	(0.00374)	(0.0138)	(0.0337)
Age in days (Right)	0.0236***	0.0243***	0.178***	0.293***	-0.0736***	-0.0740***	0.234***	0.493***
	(0.00372)	(0.00370)	(0.0142)	(0.0355)	(0.00529)	(0.00529)	(0.0195)	(0.0476)
Age in days^2 (Left)			-0.000112***	-0.000217***			-0.000189***	-0.000158
			(1.33e-05)	(7.98e-05)			(1.83e-05)	(0.000107)
Age in days^2 (Right)			1.41e-05	-0.000170			-4.29e-05*	-0.000993***
			(1.89e-05)	(0.000113)			(2.58e-05)	(0.000151)
Age in days^3 (Left)				-9.52e-08				2.88e-08
				(7.18e-08)				(9.62e-08)
Age in days^3 (Right)				3.59e-07***				8.09e-07***
				(1.01e-07)				(1.36e-07)
Birthday indicator		60.15***	68.92***	74.08***		-35.61*	-14.92	1.508
		(14.95)	(14.41)	(14.41)		(21.37)	(19.72)	(19.33)
Constant	205.2***	205.2***	195.2***	193.3***	349.1***	349.1***	332.3***	332.8***
	(1.111)	(1.105)	(1.594)	(2.121)	(1.580)	(1.579)	(2.182)	(2.845)
Bandwidth	730	730	730	730	730	730	730	730
R-squared	0.284	0.292	0.348	0.355	0.712	0.712	0.757	0.769
Mean crime freq.	207.5	207.5	207.5	207.5	363.3	363.3	363.3	363.3
Δ at 18	-10.6%	-10.8%	10.2%	-11.8%	-0.5%	-0.4%	-1.5%	-6.1%

	Violent Crimes			Property Crimes		
	Conventional (1)	Robust (2)	Epanechikov (3)	Conventional (4)	Robust (5)	Epanechikov (6)
B. Local Nonparametric Estimates						
RD at 18	-21.472***	-21.331***	-21.561***	-27.858***	-29.894***	-27.358***
	(2.057)	(2.478)	(2.042)	(4.709)	(4.965)	(4.687)
Bandwidth	501.237	501.237	467.192	183.314	183.314	173.05
Observations	1,003	1,003	935	367	367	347
Mean crime freq.	198.4	198.4	197.9	338.7	338.7	337.9
Kernel	Triangular	Triangular	Epanechikov	Triangular	Triangular	Epanechikov
Δ at 18	-10.8 %	-10.8%	-10.9%	-8.2%	-8.8%	-8.1%

Notes: Panel A presents results from global polynomial estimates. Conventional standard errors are reported across all specifications. Columns (1), (2), (5) and (6) present global first order polynomial (linear) results. Columns (3) and (7) present global second order polynomial results. Columns (4) and (8) present global third order polynomial results. Dummy variables control for birthday effects are included in Columns (2)-(4) and (6)-(8). Panel B presents results from local nonparametric estimates with the triangular and Epanechikov kernels, as well as data-driven MSE-optimal bandwidth selection methods. Columns (1) and (4) report conventional RD estimates. Columns (2) and (5) report bias-correction estimates with robust standard errors following Calonico, Cattaneo, and Titiunik (2014) and Calonico, Cattaneo, and Farrell (2017). Columns (1) and (4) report conventional RD estimates using the Epanechikov kernel. Entries in the first rows of each panel (labeled "RD at 18") are estimated regression discontinuities at age 18, measured as a decline in total daily crime counts throughout the sample period. Mean crime freq. is the baseline total average daily crime counts committed by youth just below age 18. Entries in the last rows of each panel (labeled "Δ at 18") are estimated regression discontinuities at age 18 measured as a percentage change in crime. This is calculated as the discontinuous drop in crime counts at age 18 divided by the baseline mean daily crime counts committed by youths just below age 18. In all models *, ** and *** denote statistical significance at the 10%, 5% and 1% levels, respectively.

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Table 3: Estimated Discontinuities in Violent Crimes at Age 18 by Race

	White		Hispanic		Black		Asian	
	RD at 18 (1)	Δ at 18 (2)	RD at 18 (3)	Δ at 18 (4)	RD at 18 (5)	Δ at 18 (6)	RD at 18 (7)	Δ at 18 (8)
A. Global Polynomial Estimates								
Linear	-12.20***	-20.9%	-5.506***	-6.4%	-1.133	-2.4%	-1.614***	-26.0%
	(0.784)		(1.006)		(0.729)		(0.239)	
2nd order polynomial	-10.82***	-18.6%	-6.502***	-7.5%	-0.750	-1.6%	-0.981***	-15.8%
	(1.157)		(1.488)		(1.089)		(0.359)	
3rd order polynomial	-11.45***	-19.7%	-7.980***	-9.3%	-2.186	-4.6%	-1.033**	-16.7%
	(1.548)		(1.985)		(1.452)		(0.480)	
Observations	1,461		1,461		1,461		1,461	
Bandwidth	730		730		730		730	
Mean crime freq.	58.3		86.3		48.0		6.2	
B. Local Linear Estimates								
Conventional	-11.243***	-21.1%	-5.584***	-6.5%	-0.85	-1.9%	-1.241***	-23.4%
	(1.015)		(1.185)		(0.837)		(0.285)	
Robust	-10.750***	-20.2%	-5.704***	-6.6%	-0.465	-1.0%	-1.117***	-21.1%
	(1.063)		(1.426)		(0.924)		(0.321)	
Epanechikov	-11.292***	-21.3%	-5.484***	-6.4%	-0.644	-1.4%	-1.257***	-23.8%
	(1.019)		(1.154)		(0.855)		(0.288)	
Observations	1,013		1,271		1,211		1,109	
Bandwidth	506.6		635.7		605.2		554.0	
Mean crime freq.	53.2		86.2		45.7		5.3	

Notes: Columns (1) and (2) present RD estimates for white offenders. Columns (3) and (4) present RD estimates for Hispanic offenders. Columns (5) and (6) present RD estimates for black offenders. Columns (7) and (8) present RD estimates for Asian offenders. Racial determinations are made by the arresting law enforcement agencies. Columns (1), (3), (5) and (7) are estimated regression discontinuities at age 18, measured as the decline in total daily crime counts throughout the sample period. Columns (2), (4), (6) and (8) are estimated regression discontinuities at age 18 measured as a percentage change in crime, calculated as the discontinuous drop in crime counts at age 18 divided by the baseline mean daily crime counts committed by youth just below age 18. Mean crime freq. is the baseline total average daily crime counts committed by youth just below age 18. Panel A presents results from global polynomial estimates. Conventional standard errors are reported across all specifications. The first, second, and third rows present global first, second, and third order polynomial results, respectively. Panel B presents results from local nonparametric estimates with data-driven MSE-optimal bandwidth selection methods. The top row presents conventional RD estimates with triangular kernel. The second row present bias-correction estimates with robust standard errors with triangular kernel. The third row presents conventional estimates with Epanechikov kernel. The reported bandwidth, observations, and mean crime frequency are calculated using the optimal bandwidth under the triangular kernel. In all models *, ** and *** denote statistical significance at the 10%, 5% and 1% levels, respectively.

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Table 4: Estimated Discontinuities in Violent Crimes at Age 18 by Gender

	Male		Female	
	RD at 18	Δ at 18	RD at 18	Δ at 18
	(1)	(2)	(3)	(4)
A. Global Polynomial Estimates				
Linear	-14.78***	-9.0%	-7.575***	-17.4%
	(1.413)		(0.635)	
Second order polynomial	-13.57***	-8.3%	-7.536***	-17.3%
	(2.053)		(0.947)	
Third order polynomial	-16.40***	-10.0%	-8.002***	-18.3%
	(2.732)		(1.266)	
Observations	1,461		1,461	
Bandwidth	730		730	
Mean crime freq.	163.9		43.65	
B. Local Linear Estimates				
Conventional	-13.921***	-8.6%	-7.733***	-21.9%
	(1.986)		(0.869)	
Robust	-13.009***	-8.0%	-7.793***	-22.1%
	(2.053)		(1.046)	
Epanechikov	-13.917***	-8.6%	-7.776***	-22.2%
	(1.976)		(0.874)	
Observations	941	941	949	949
Bandwidth	470.3	470.3	474.7	474.7
Mean crime freq.	162.7	162.7	35.3	35.3

Notes: Columns (1) and (2) present RD estimates for male offenders. Columns (3) and (4) present RD estimates for female offenders. Columns (1) and (3) are estimated regression discontinuities at age 18, measured as the decline in total daily crime counts throughout the sample period. Columns (2) and (4) are estimated regression discontinuities at age 18 measured as a percentage change in crime, calculated as the discontinuous drop in crime counts at age 18 divided by the baseline mean daily crime counts committed by youth just below age 18. Mean crime freq. is the baseline total average daily crime counts committed by youth just below age 18. Panel A presents results from global polynomial estimates. Conventional standard errors are reported across all specifications. The first, second, and third rows present global first, second, and third order polynomial results, respectively. Panel B presents results from local nonparametric estimates with data-driven MSE-optimal bandwidth selection methods. The top row presents conventional RD estimates with triangular kernel. The second row present bias-correction estimates with robust standard errors with triangular kernel. The third row presents conventional estimates with Epanechikov kernel. The reported bandwidth, observations, and mean crime frequency are calculated using the optimal bandwidth under the triangular kernel. In all models *, ** and *** denote statistical significance at the 10%, 5% and 1% levels, respectively.

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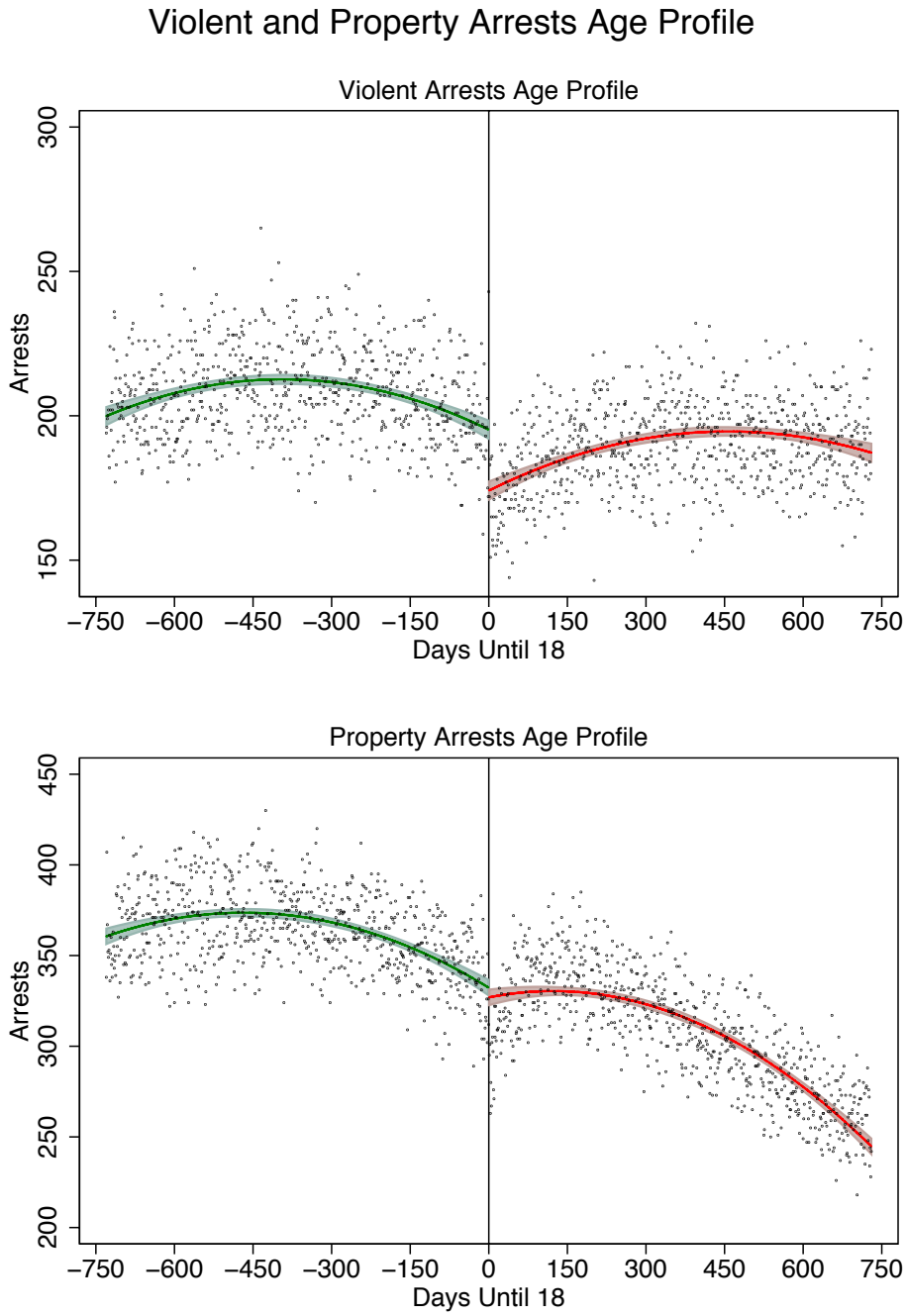
Figure 1: Violent and Property Crime by Age



Notes: The x-axis is days relative to the age of majority. The y-axis reports the incidence of violent and property crime committed by a precise age group. The fitted curves (with 95% confidence intervals) superimposed over the daily crime counts are from a second order polynomial in age fully interacted with an indicator variable for having reached the age of majority.

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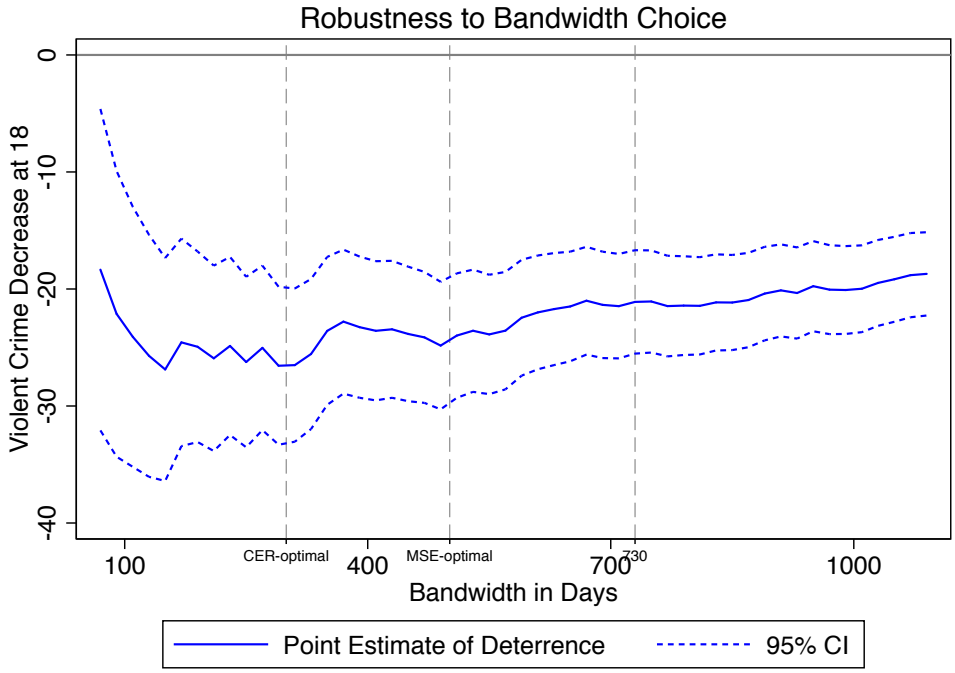
Figure 2: Robustness to Bandwidth Choice



Notes: The x-axis presents a variety of bandwidths used in the RD analysis ranging from approximately two months to three years. The estimates of the decrease in violent crime at age 18 are from a second order polynomial fully interacted with an indicator variable for being over 18. The figure includes three vertical lines denoting the Calonico, Cattaneo, Titiunik (CCT) optimal bandwidth, the Coverage Error Rate (CER) optimal bandwidth, and the ±730 day bandwidth that is used as a baseline throughout the paper. The solid line is the point estimate; the short dashed lines indicate the 95% confidence interval.

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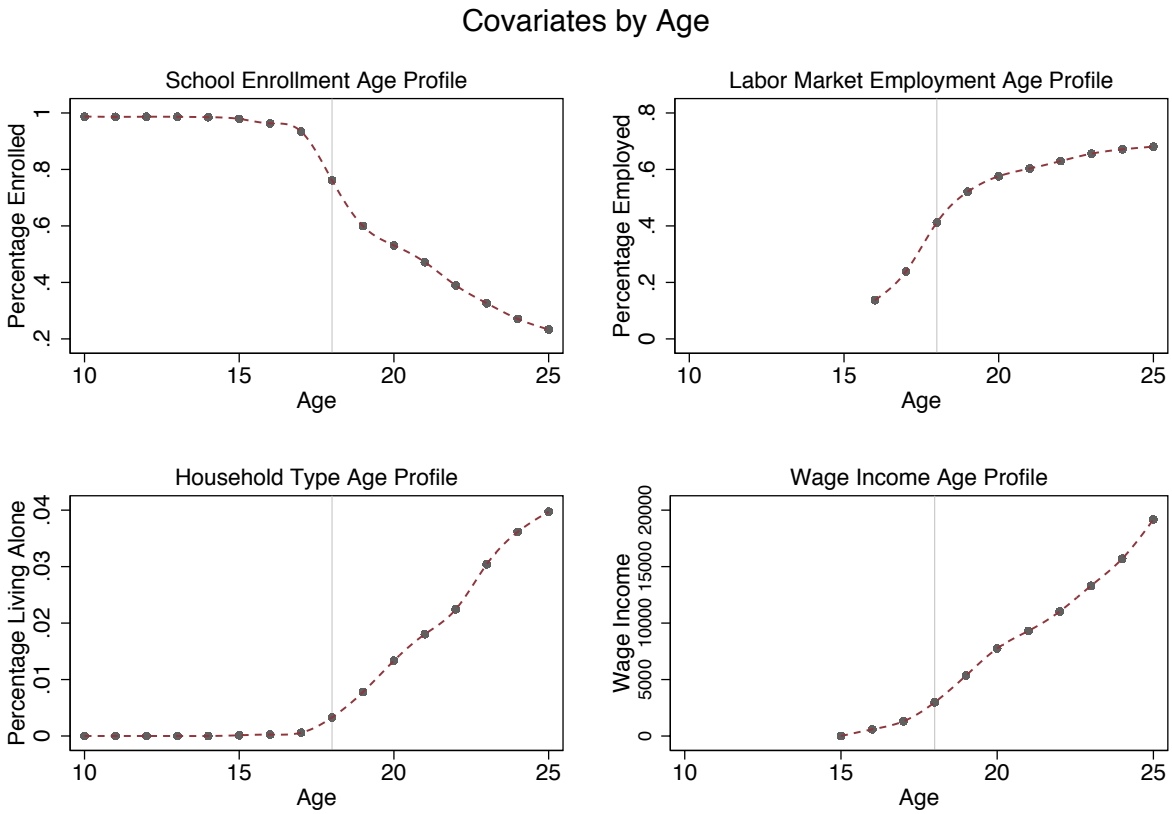
Figure 3: Covariates Smooth Around Age 18



Notes: These figures are plotted using data from ACS 2000-2006 and Decennial Census. The top left panel plots the probability of school enrollment at each age. The top right panel plots the probability of being employed at each age. The bottom left panel plots the probability of an individual living alone at each age. The bottom right panel plots the average wage income at each age.

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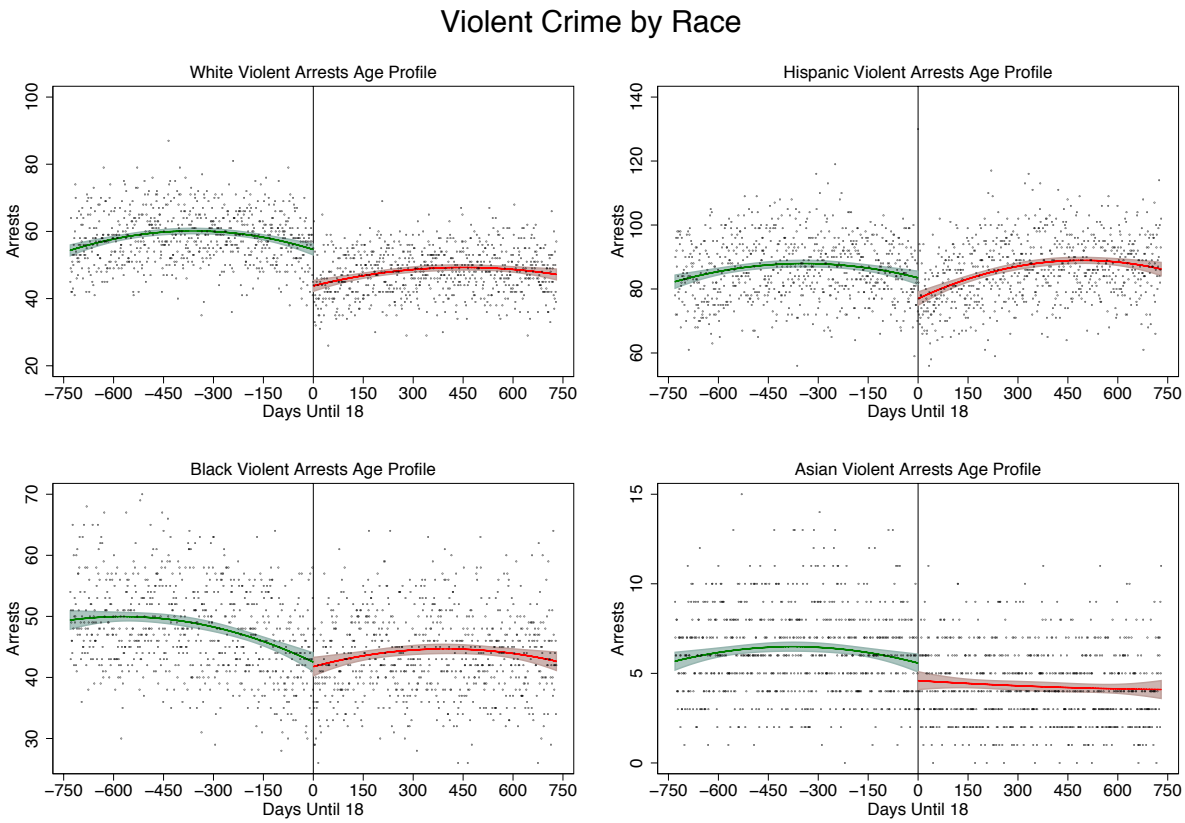
Figure 4: Violent Crime by Race



Notes: The x-axis is days relative to the age of majority. The y-axis reports the incidence of violent crime committed by a precise age group. Racial determinations are made by the arresting law enforcement agencies. The fitted curves (with 95% confidence intervals) superimposed over the daily crime counts are from a second order polynomial in age fully interacted with an indicator variable for having reached the age of majority.

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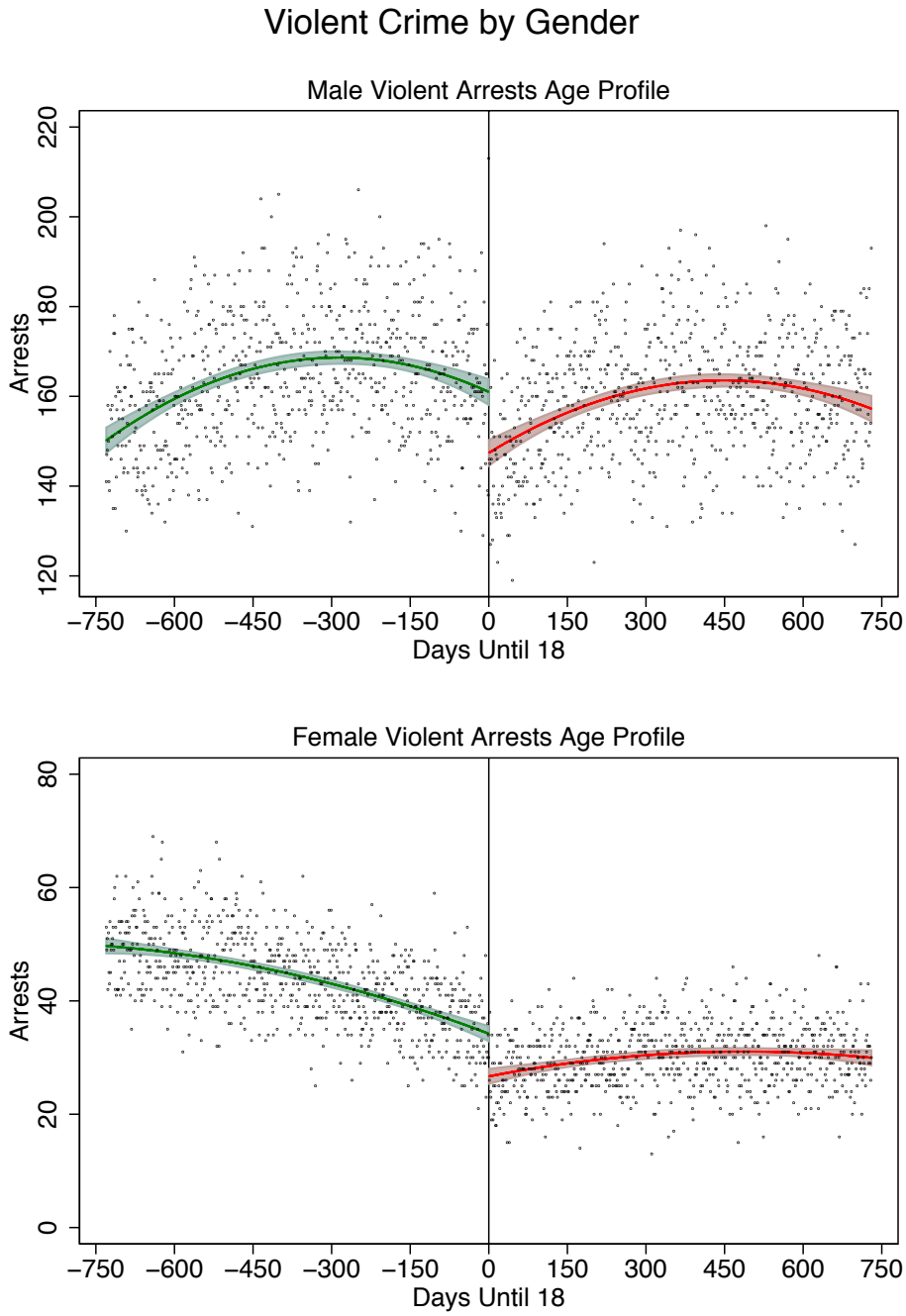
Figure 5: Violent Crime by Gender



Notes: The x-axis is days relative to the age of majority. The y-axis reports the incidence of violent crime committed by a precise age group. The fitted curves (with 95% confidence intervals) superimposed over the daily crime counts are from a second order polynomial in age fully interacted with an indicator variable for having reached the age of majority.

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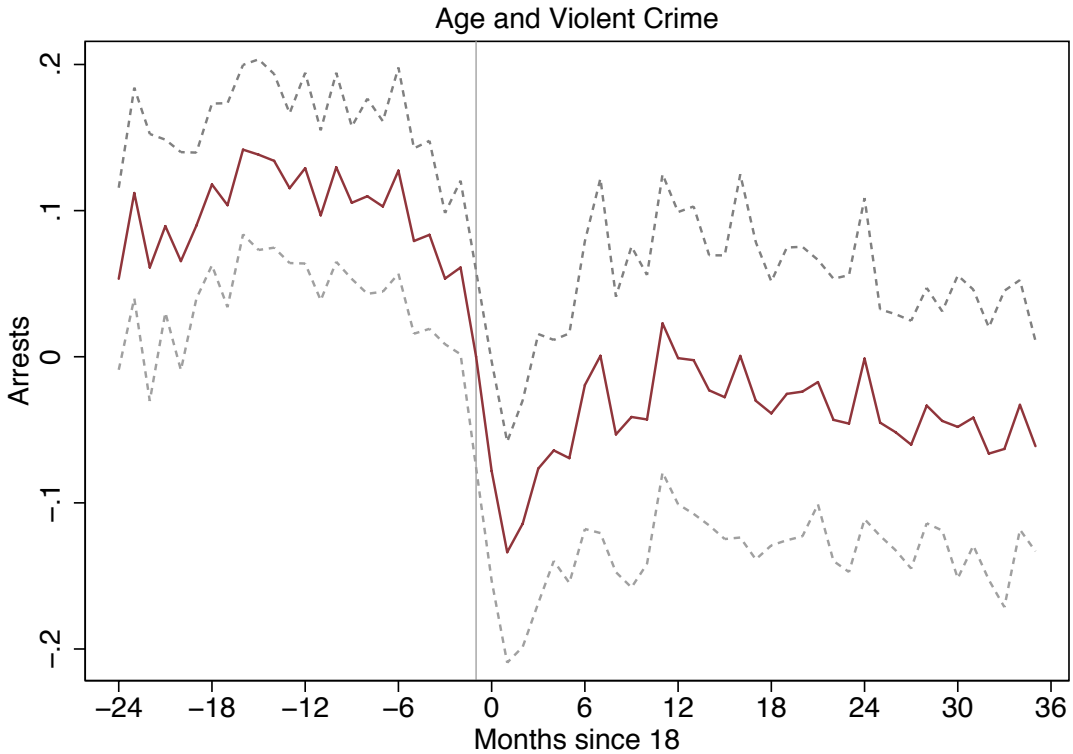
Figure 6: Nonparametric Event Study: Crime by Age



Notes: The months on the x-axis are defined relative to the age of majority. The y-axis reports the incidence of violent crime relative to violent crime in the baseline month, which is the last month before individuals reach the age of majority. Each unit of observation is a violent crime count, committed by a precise age group, in a calendar month and year in a particular county. The mean level of violent crime in the baseline period is 1.25. The point estimates in the figure represent the effects of event time, with the month prior to the age of majority normalized to zero. Standard errors are clustered at the county level. The dashed lines represent 95% confidence intervals.

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